CSE-400 Scribe - L15

L15_L17.pdf
PDF

Generate a lecture scribe intended strictly as exam-oriented reference material for CSE 400. Use only the provided context (lecture slides/pdfs and the explicitly relevant textbook parts). Treat this context as authoritative and complete. Do not use or introduce outside knowledge.

When processing handwritten notes, read them carefully and conserve their meanings. Interpret symbols, notation, and equations with precision. Resolve ambiguities strictly using consistency with the lecture slides and textbook content. Draw figures/diagrams when needed. Do not guess or hallucinate unclear material.

Write in a formal, academic tone suitable for closed-notes exam revision. Preserve the lecture's structure, ordering, notation, terminology, and scope exactly. While writing the scribe, reason step by step through the material exactly as presented in the context, and include that reasoning explicitly in the scribe.

Emphasize clarity of dependencies between assumptions, definitions, and results without introducing new explanations.

Focus more on: 1-State and explain all definitions, notation, and assumptions explicitly, 2-Present theorems, propositions, and results exactly as given, 3-Present proofs and derivations step by step, preserving all logical dependencies and, 4-Present worked examples step by step, showing all intermediate steps exactly as taught.

Content constraints: 1-Include only material present in the provided context, 2-Do not introduce new examples, interpretations, intuitions, alternative

arguments, or additional explanations, 3-Do not omit conditions, assumptions, or logical steps present in the source material and, 4-Do not simplify, generalize, or paraphrase beyond faithful reconstruction.

Organization Note: Structure the scribe with clear sections and headings reflecting the lecture flow so that a student relying solely on this document can accurately reconstruct definitions, derivations, proofs, and example solutions for exam preparation.

CSE 400 — Lecture Scribe

Joint Random Variables and Joint Distributions

(Lectures L15–L17)

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1. Motivation: Why Study Joint Random Variables

In many real-world systems, multiple random variables must be analyzed simultaneously. The behavior of one variable often depends on another.

Example 1: Smart Transportation

Random variables:

- X = Vehicle Speed
- Y = Traffic Density

Question:
Can speed be high when traffic density is also high?

This indicates that the two variables may be **correlated random variables**.

Applications:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Random variables:

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Question:
If SNR is high, will throughput always be high?

Throughput depends on SNR, but may also depend on other conditions such as network congestion.

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Additional Examples

- Weather prediction
  X = Rainfall, Y = Temperature
- Drone delivery
  X = Wind Speed, Y = Delivery Time
- Rolling two dice
  Requires joint probability calculations.

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2. Discrete Case — Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X \in \{x_1, x_2, \dots, x_n\}$$

$$Y \in \{y_1, y_2, \dots, y_m\}$$

The ordered pair (X, Y) takes values in the **product set**

$$\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$$

Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

This represents the probability that the **two outcomes occur simultaneously**.

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3. Joint PMF Table Representation

The joint PMF can be organized in a table.

$X \backslash Y$	y_1	y_2	...	y_j	...	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...	$p(x_1, y_j)$	...	$p(x_1, y_m$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...	$p(x_2, y_j)$	...	$p(x_2, y_m$
...	...	...	...	...	...	...
x_i	$p(x_i, y_1)$	$p(x_i, y_2)$	...	$p(x_i, y_j)$	...	$p(x_i, y_m$
...	...	...	...	...	...	...
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	...	$p(x_n, y_j)$	...	$p(x_n, y_m$

Key Properties

$$0 \leqslant p(x_i, y_j) \leqslant 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

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4. Example: Rolling Two Dice

Experiment

Roll two fair dice.

Let

$$X = \text{value on the first die}$$

$$T = \text{total (sum) of both dice}$$

Each outcome (i, j) has probability

$$\frac{1}{36}$$

Construct the **joint probability table of** (X, T).

Joint Probability Table

$X \backslash T$	**2**	**3**	**4**	**5**	**6**	**7**
1	1/36	1/36	1/36	1/36	1/36	1/36
2	0	1/36	1/36	1/36	1/36	1/36
3	0	0	1/36	1/36	1/36	1/36
4	0	0	0	1/36	1/36	1/36
5	0	0	0	0	1/36	1/36
6	0	0	0	0	0	1/36

Observations

- Entries are either 0 or $1/36$.
- X and T are **not independent**.

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5. Continuous Case — Joint PDF

The continuous case is analogous to the discrete case.

Correspondence:

Discrete	**Continuous**
Values	Intervals

Discrete	Continuous
Joint PMF	Joint PDF
Sums	Double integrals

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) lies in the **product region**

$$[a, b] \times [c, d]$$

Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** of X and Y if

$$P((X, Y) \text{ lies in a small rectangle of area } dx\,dy)$$
$$\approx f(x, y)\,dx\,dy$$

Key Properties

$$f(x, y) \geqslant 0$$
$$\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$$

Note:

The joint PDF $f(x, y)$ may take values **greater than 1**, because it is a **density**, not a probability.

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6. Worked Example

Given

$$f(x, y) = 4xy$$

for

$$0 \leqslant x \leqslant 1, \quad 0 \leqslant y \leqslant 1$$

Thus (X, Y) lies in the **unit square**

$$[0, 1]^2$$

Tasks

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$.

Example Solution

Step 1: Check Validity (Normalization)

Compute

$$\int_0^1 \int_0^1 4xy\, dx\, dy$$

First integrate with respect to x:

$$= \int_0^1 \left(\int_0^1 4xy\, dx \right) dy$$

$$= \int_0^1 2y\, dy$$

$$= [y^2]_0^1$$

$$= 1$$

Thus

$$f(x, y) \geqslant 0$$

and total probability equals 1.

Therefore $f(x, y)$ is a **valid joint PDF**.

Step 2: Compute $P(A)$

$$A = \{X < 0.5,\ Y > 0.5\}$$

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy\, dy\, dx$$

First integrate with respect to y:

$$= \int_0^{0.5} [2xy^2]_{0.5}^{1} dx$$

$$= \int_0^{0.5} 2x(1 - 0.25)\, dx$$

$$= \int_0^{0.5} \frac{3}{2} x\, dx$$

$$= \frac{3}{16}$$

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7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation

$$X \leqslant x, \quad Y \leqslant y$$

means the event

$$\{X \leqslant x \text{ and } Y \leqslant y\}$$

Definition

The **joint cumulative distribution function** is

$$F(x, y) = P(X \leqslant x, Y \leqslant y)$$

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8. Joint CDF — Continuous Case

If X and Y have joint density $f(x, y)$ over

$$[a, b] \times [c, d]$$

then

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recovering the Joint PDF

Since two variables are involved, use **partial derivatives**:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

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9. Joint CDF — Discrete Case

If X and Y are discrete with joint PMF $p(x_i, y_j)$,

then

$$F(x, y) = \sum_{x_i \leqslant x} \sum_{y_j \leqslant y} p(x_i, y_j)$$

Interpretation

To compute $F(x, y)$:

- Add probabilities in the table
- Include rows where $x_i \leqslant x$
- Include columns where $y_j \leqslant y$

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10. Properties of the Joint CDF

The joint CDF $F(x, y)$ must satisfy:

1. Non-decreasing

$$F(x, y) \text{ is non-decreasing.}$$

If x or y increases, $F(x, y)$ must stay the same or increase.

2. Lower-Left Boundary

$$F(x, y) = 0$$

at the lower-left of the joint range.

If the corner is $(-\infty, -\infty)$,

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

3. Upper-Right Boundary

$$F(x, y) = 1$$

at the upper-right of the joint range.

If the corner is (∞, ∞),

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$